

Worksheet - Sound and Hearing

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the exported image in the marked space.

Part 1 - Sound waves

Task 1: Investigating pitch

Set the mode to "continuous pulse" and choose the "sound waves" display. Set the volume to 50%. Now investigate a low pitch (e.g. 200 Hz) and a high pitch (e.g. 900 Hz). Export both states in the microphone measurement and then compare the spacing of the sound waves.

Low pitch
Insert here
High pitch
Insert here

Task 2: Investigating volume

Keep the pitch constant (e.g. 440 Hz) and compare a low volume (e.g. 20%) with a high volume (e.g. 90%). Export both states in the microphone measurement and then compare the spacing of the sound waves.

Low volume
Insert here
High volume
Insert here

Task 3: Transmission of sound waves

Switch to the "particles" display and read the additional explanation. Explain how a sound wave is transmitted through air and describe the role of sound pressure in perceiving volume. Export the microphone measurement at a pitch of 440 Hz and a volume of 65%.

Particle model
Insert here
